the guaranteed payoff paradox

The guaranteed payoff paradox isn't described as such in the textbook, but you've already seen on a homework assignment. I'll say a little bit more about it here, though.

Imagine that you are offered this decision.

	P = .01 ticket # 1 is drawn	P = .10 # 2 – 11 is drawn	P = .89 #12 – 100 is drawn
lottery A	$100	$100	$100
lottery B	$0	$500	$100

As you know from the homework assignment, the expected value for lottery B is higher than the expected value for lottery A.

EV(lottery A) = (.01)(100) + (.10)(100) + (.89)(100) = 1 + 10 + 89 = $100

EV(lottery B) = (.01)(100) + (.10)(100) + (.89)(100) = 0 + 50 + 89 = **$139**

Let's assume that, for this decision problem, most people would follow the expected value strategy and select lottery B. Some people, however, would choose lottery A.

If we proportionally increase the outcomes, in the following way, the expected value of lottery B will always be higher.

	P = .01 ticket # 1 is drawn	P = .10 # 2 – 11 is drawn	P = .89 #12 – 100 is drawn
lottery A	$10,000	$10,000	$10,000
lottery B	$0	$50,000	$10,000

EV(lottery A) = $10,000

EV(lottery B) = **$13,900**

	P = .01 ticket # 1 is drawn	**P** = .10 # 2 – 11 is drawn	**P** = .89 #12 – 100 is drawn
lottery A	$1,000,000	$1,000,000	$1,000,000
lottery B	$0	$5,000,000	$1,000,000

EV(lottery A) = $1,000,000

EV(lottery B) = **$1,390,000**

	P = .01 ticket # 1 is drawn	**P** = .10 # 2 – 11 is drawn	**P** = .89 #12 – 100 is drawn
lottery A	$10,000,000	$10,000,000	$10,000,000
lottery B	$0	$50,000,000	$10,000,000

EV(lottery A) = $10,000,000

EV(lottery B) = **$13,900,000**

	P = .01 ticket # 1 is drawn	**P** = .10 # 2 – 11 is drawn	**P** = .89 #12 – 100 is drawn
lottery A	$100,000,000 (i.e., $100M)	$100,000,000	$100,000,000
lottery B	$0	$500,000,000	$100,000,000

EV(lottery A) = $100,000,000

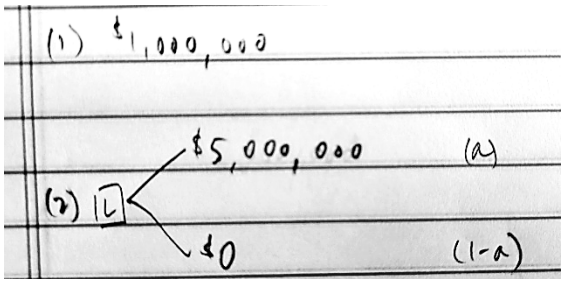
EV(lottery B) = **$139,000,000**

When offered these lotteries, almost all people, at some point, will choose lottery A because it offers the guaranteed payoff. When they do so, they are rejecting the expected value strategy. But interestingly, we also can't explain their decision with the expected utility strategy. Let's examine why that is.

Everyone is going to rank the outcomes in the same order. So, if we use the decision problem with the $0, $1,000,000, and $5,000,000 outcomes, we start with this:

$5,000,000	1
$1,000,000	__
$0	0

To determine the utility value for the $1,000,000, we set up the two options like this:



What probability of winning the lottery would make you indifferent between (1) and (2)? Well, if you are choosing lottery A, then the probability would, basically, have to be 100%. After all, a 1% chance of getting $0 is why you are choosing lottery A over lottery B. This is odd. I would choose lottery A, but I think that I value $5 million more than I value $1 million. In other words, I don't think that these really are my utility values:

$5,000,000	1
$1,000,000	1
$0	0

I know that when amounts of money get sufficiently large, I don't value a small increase very much or maybe at all. For instance, I have essentially the same preference for $3,000,000 and $3,000,010 ($10 more). But I think my preference for $5 million should be distinguishable from my preference for $1 million. The best available method for determining utility values, however, indicates that—at least in this decision problem with $0 as the least preferred outcome—they are not.